

Table 1: The expected sojourn time for the core parallel computational phase from our three interpolation approximations as a function of the system load $\rho = \gamma/\mu^*$ and the heterogeneity ratio $r > 1$ in comparison against simulation, with the lowest error highlighted in bold. All three approximations are exact when $r = 1$. Each simulation entry is the grand mean over 10 independent replications of 20 million jobs each, with a warm-up period of 500 thousand jobs discarded per replication; the reported 95% confidence interval is the Student- t interval across the 10 replica means.

ρ	r	Simulation	$\mathbb{E}[T_{FJ}]$ in		$\mathbb{E}[T_{FJ}^{(0)}]$ in		$\mathbb{E}[T_{FJ}^{(1)}]$ in	
		$\mathbb{E}[T_{\text{sim}}] \pm 95\% \text{ CI}$	$\mathbb{E}[T]$	Err	$\mathbb{E}[T]$	Err	$\mathbb{E}[T]$	Err
0.4	2	1.814 ± 0.001	1.824	+0.57%	1.834	+1.08%	1.825	+0.59%
	4	1.702 ± 0.001	1.704	+0.12%	1.717	+0.87%	1.705	+0.16%
	8	1.675 ± 0.001	1.676	+0.02%	1.681	+0.31%	1.676	+0.03%
0.8	2	5.080 ± 0.008	5.101	+0.41%	5.168	+1.73%	5.151	+1.38%
	4	5.014 ± 0.008	5.016	+0.05%	5.050	+0.72%	5.026	+0.24%
	8	5.003 ± 0.008	5.003	+0.01%	5.014	+0.22%	5.005	+0.04%
0.9	2	10.049 ± 0.039	10.063	+0.13%	10.170	+1.20%	10.150	+1.00%
	4	10.010 ± 0.039	10.009	−0.02%	10.050	+0.40%	10.023	+0.12%
	8	10.004 ± 0.039	10.002	−0.03%	10.014	+0.09%	10.004	−0.01%
0.95	2	20.073 ± 0.149	20.036	−0.19%	20.174	+0.50%	20.153	+0.40%
	4	20.052 ± 0.149	20.005	−0.24%	20.050	−0.01%	20.021	−0.15%
	8	20.049 ± 0.149	20.001	−0.24%	20.014	−0.17%	20.003	−0.23%